

REC-CIS

GE23131-Programming Using C-2024

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Status	Finished
Started	Monday, 13 January 2025, 2:03 PM
Completed	Monday, 13 January 2025, 2:08 PM
Duration	5 mins 12 secs

Question 1

Correct

Marked out of 1.00

Flag question

Given an array of integers, reverse the given array in place using an index and loop rather than a built-in function.

Example

`arr = [1, 3, 2, 4, 5]`

Return the array `[5, 4, 2, 3, 1]` which is the reverse of the input array.

Function Description

Complete the function `reverseArray` in the editor below.

`reverseArray` has the following parameter(s):

`int arr[n]`: an array of integers

Return

`int[n]`: the array in reverse order

Constraints

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$$1 \leq n \leq 100$$

$$0 < arr[i] \leq 100$$

Input Format For Custom Testing

The first line contains an integer, n , the number of elements in arr .

Each line i of the n subsequent lines (where $0 \leq i < n$) contains an integer, $arr[i]$.

Sample Case 0**Sample Input For Custom Testing**

5

1

3

2

4

5

Sample Output

5

4

2

3

1

Explanation

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The input array is [1, 3, 2, 4, 5], so the reverse of the input array is [5, 4, 2, 3, 1].

Sample Case 1**Sample Input For Custom Testing**

4

17

10

21

45

Sample Output

45

21

10

17

Explanation

The input array is [17, 10, 21, 45], so the reverse of the input array is [45, 21, 10, 17].

Answer: (penalty regime: 0 %)

Reset answer

```
1 /*  
2  * Complete the 'reverseArray' function below.  
3  *  
4  * The function is expected to return an INTEGER_ARRAY.
```

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```
5  * The function accepts INTEGER_ARRAY arr as parameter.
6  */
7
8  /*
9  * To return the integer array from the function, you should:
10 *   - Store the size of the array to be returned in the result_count variable
11 *   - Allocate the array statically or dynamically
12 *
13 * For example,
14 * int* return_integer_array_using_static_allocation(int* result_count) {
15 *     *result_count = 5;
16 *
17 *     static int a[5] = {1, 2, 3, 4, 5};
18 *
19 *     return a;
20 * }
21 *
22 * int* return_integer_array_using_dynamic_allocation(int* result_count) {
23 *     *result_count = 5;
24 *
25 *     int *a = malloc(5 * sizeof(int));
26 *
27 *     for (int i = 0; i < 5; i++) {
28 *         *(a + i) = i + 1;
29 *     }
30 *
31 *     return a;
32 * }
33 *
34 */
35 int* reverseArray(int arr_count, int *arr, int *result_count) {
36     *result_count=arr_count;
37     static int rev[100];
38     int i = 0;
```

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```

37 static int rev[100];
38 int i,j=0;
39 for(i=arr_count-1;i>=0;i--)
40     rev[j++]=arr[i];
41 return rev;
42
43 }
44

```

	Test	Expected	Got	
✓	<pre> int arr[] = {1, 3, 2, 4, 5}; int result_count; int* result = reverseArray(5, arr, &result_count); for (int i = 0; i < result_count; i++) printf("%d\n", *(result + i)); </pre>	5 4 2 3 1	5 4 2 3 1	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 1.00

An automated cutting machine is used to cut rods into segments. The cutting machine can only hold a rod of *minLength* or more, and it can only make one cut at a time. Given the array *lengths[]* representing the desired lengths of each segment, determine if it is possible to make the necessary cuts using this machine. The rod is marked into *lengths* already, in the order given.



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Example

$$n = 3$$

$$\text{lengths} = [4, 3, 2]$$

$$\text{minLength} = 7$$

The rod is initially $\text{sum}(\text{lengths}) = 4 + 3 + 2 = 9$ units long. First cut off the segment of length $4 + 3 = 7$ leaving a rod $9 - 7 = 2$. Then check that the length 7 rod can be cut into segments of lengths 4 and 3. Since 7 is greater than or equal to $\text{minLength} = 7$, the final cut can be made. Return "Possible".

Example

$$n = 3$$

$$\text{lengths} = [4, 2, 3]$$

$$\text{minLength} = 7$$

The rod is initially $\text{sum}(\text{lengths}) = 4 + 2 + 3 = 9$ units long. In this case, the initial cut can be of length 4 or $4 + 2 = 6$. Regardless of the length of the first cut, the remaining piece will be shorter than minLength . Because $n - 1 = 2$ cuts cannot be made, the answer is "Impossible".

Function Description

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Complete the function *cutThemAll* in the editor below.

cutThemAll has the following parameter(s):

int lengths[n]: the lengths of the segments, in order

int minLength: the minimum length the machine can accept

Returns

string: "Possible" if all $n-1$ cuts can be made. Otherwise, return the string "Impossible".

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq t \leq 10^9$
- $1 \leq \text{lengths}[i] \leq 10^9$
- The sum of the elements of *lengths* equals the uncut rod length.

Input Format For Custom Testing

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The first line contains an integer, n , the number of elements in *lengths*.

Each line i of the n subsequent lines (where $0 \leq i < n$) contains an integer, *lengths*[i].

The next line contains an integer, *minLength*, the minimum length accepted by the machine.

Sample Case 0**Sample Input For Custom Testing**

STDIN Function

4 → lengths[] size $n = 4$

3 → lengths[] = [3, 5, 4, 3]

5

4

3

9 → minLength = 9

Sample Output

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Possible

Explanation

The uncut rod is $3 + 5 + 4 + 3 = 15$ units long. Cut the rod into lengths of $3 + 5 + 4 = 12$ and 3. Then cut the 12 unit piece into lengths 3 and $5 + 4 = 9$. The remaining segment is $5 + 4 = 9$ units and that is long enough to make the final cut.

Sample Case 1**Sample Input For Custom Testing**

STDIN	Function
-----	-----
3	→ lengths[] size n = 3
5	→ lengths[] = [5, 6, 2]
6	
2	
12	→ minLength= 12

Sample Output

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Impossible

Explanation

The uncut rod is $5 + 6 + 2 = 13$ units long. After making either cut, the rod will be too short to make the second cut.

Answer: (penalty regime: 0 %)

Reset answer

```
1 /*
2  * Complete the 'cutThemAll' function below.
3  *
4  * The function is expected to return a STRING.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER_ARRAY lengths
7  * 2. LONG_INTEGER minLength
8  */
9
10 /*
11  * To return the string from the function, you should either do static allocation or dynamic allocation.
12  *
13  * For example,
14  * char* return_string_using_static_allocation() {
15  *     static char s[] = "static allocation of string";
16  *
17  *     return s;
```

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```
18 * }
19 *
20 * char* return_string_using_dynamic_allocation() {
21 *     char* s = malloc(100 * sizeof(char));
22 *
23 *     s = "dynamic allocation of string";
24 *
25 *     return s;
26 * }
27 *
28 */
29 char* cutThemAll(int lengths_count, long *lengths, long minLength) {
30     int s=0;
31     for(int i=0;i<lengths_count-1;i++){
32         s+= *(lengths+i);
33     }
34     if(s>=minLength){
35         return "Possible";
36     }
37     else{
38         return "Impossible";
39     }
40 }
41 }
42 }
```

Test	Expected	Got

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```
32     s = (char*)malloc(sizeof(char) * minLength);  
33 }  
34 if(s != NULL){  
35     return "Possible";  
36 }  
37 else{  
38     return "Impossible";  
39 }  
40 }  
41 }  
42 }
```

	Test	Expected	Got	
✓	long lengths[] = {3, 5, 4, 3}; printf("%s", cutThemAll(4, lengths, 9))	Possible	Possible	✓
✓	long lengths[] = {5, 6, 2}; printf("%s", cutThemAll(3, lengths, 12))	Impossible	Impossible	✓

Passed all tests! ✓

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